Architecture 1001: x86-64 Assembly ArrayLocalVariable2.c with GCC

Xeno Kovah – 2021
xeno@darkmentor.com

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ArrayLocalVariable2.c

Zero-initializing the array

```
short main(){
    int a;
    short b[64] = {0};
    a = 0x5eaf00d;
    b[1] = (short)a;
    return b[1];
}
```



```
0x000055555555129 <+0>:     endbr64
0x00005555555512d <+4>:     push   %rbp
0x00005555555512e <+5>:     mov    %rsp,%rbp
0x000055555555131 <+8>:     sub    $0x18,%rsp
0x000055555555135 <+12>:    lea    -0x90(%rbp),%rdx
0x00005555555513c <+19>:    mov    $0x0,%eax
0x000055555555141 <+24>:    mov    $0x10,%ecx
0x000055555555146 <+29>:    mov    %rdx,%rdi
0x000055555555149 <+32>:    rep stos %rax,%es:(%rdi)
0x00005555555514c <+35>:    movl   $0x5eaf00d,-0x4(%rbp)
0x000055555555153 <+42>:    mov    -0x4(%rbp),%eax
0x000055555555156 <+45>:    mov    %ax,-0x8e(%rbp)
0x00005555555515d <+52>:    movzwl -0x8e(%rbp),%eax
0x000055555555164 <+59>:    leaveq
0x000055555555165 <+60>:    retq
```

The stack looks like this at line 0x00005555`5555515d in func():

